

Intervals for S_{64} and S_{65} , the smallest Carmichael numbers with exactly 64 and 65 prime factors

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Abstract

Let S_{64} denote the smallest Carmichael number with exactly 64 distinct prime factors (the $k = 64$ entry of OEIS [A006931](#), currently unknown). We give an explicit two-sided bound. For the upper bound we exhibit a 148-digit Carmichael number N with exactly 64 prime factors, so $S_{64} \leq N$. For the lower bound we prove, by a finite exhaustion over a certified prime universe, that no 64-factor Carmichael number is smaller than 10^{145} . Together,

$$10^{145} \leq S_{64} \leq N < 10^{148},$$

so S_{64} has 146, 147, or 148 decimal digits. All arithmetic is exact. The lower bound is a reproducible computation: the admissible universe is bounded by a Korselt cofactor argument, the resulting bound is recomputed independently, and the exhaustion is carried out by three separate enumeration implementations, one of which takes its Korselt verdict at every leaf from a frozen reference oracle. The same cofactor minimum transfers to the next factor count, giving $10^{148} \leq S_{65} \leq U_{65}$ for an explicit verified 151-digit Carmichael number U_{65} , so the smallest 65-factor Carmichael number has 149, 150, or 151 digits (§5). The author is responsible for the mathematics and code; see the acknowledgements for the role of AI assistance.

1 Statement

A *Carmichael number* is a composite integer n with $b^n \equiv b \pmod{n}$ for all integers b . By Korselt's criterion, a composite integer n is Carmichael if and only if it is squarefree and $(p-1) \mid (n-1)$ for every prime $p \mid n$. Write $\omega(n)$ for the number of distinct prime factors. The sequence $S_k = \min\{n : n \text{ Carmichael}, \omega(n) = k\}$ is OEIS [A006931](#); values are known for $3 \leq k \leq 63$ (R. G. E. Pinch for the smaller k ; the $k = 36$ – 63 values from the Butler University computation, collected in J. Webster's repository cited below), and S_{64} is open as of September 2026. This note proves:

Theorem 1 (Interval). $10^{145} \leq S_{64} \leq N$, where

$$N = 866420555770562713589289732117698405485089408042974666135014101183659858491661736151818531159$$

is a 148-digit Carmichael number with exactly 64 prime factors. Consequently S_{64} has 146, 147, or 148 decimal digits, and the remaining exclusion interval for S_{64} is $[10^{145}, N)$.

The two bounds are proved in §2 and §3. We use throughout the following elementary consequence of Korselt's criterion.

Lemma 1 (Pairwise admissibility). *If $p < q$ are distinct primes dividing a Carmichael number n , then $p \nmid (q-1)$. Moreover n is odd.*

Proof. Since $q \mid n$ we have $(q-1) \mid (n-1)$, so $n \equiv 1 \pmod{d}$ for every divisor d of $q-1$. If $p \mid (q-1)$ then $n \equiv 1 \pmod{p}$; but $p \mid n$ gives $n \equiv 0 \pmod{p}$, forcing $p \mid 1$, a contradiction. Oddness: no even integer > 2 is Carmichael, since for an odd prime factor p the even number $p-1$ cannot divide the (then even) $n-1$ consistently with squarefreeness; concretely 2 cannot be one of $k \geq 3$ distinct prime factors of a Carmichael number. $\square$

2 Upper bound

The number N of Theorem 1 is the product of the 64 primes

19, 29, 31, 37, 41, 43, 47, 53, 61, 67, 71, 73, 79, 89, 97, 101, 103, 109, 113, 127,
131, 137, 139, 151, 157, 163, 167, 181, 193, 197, 199, 211, 239, 241, 251, 257, 271, 277, 281, 307,
313, 331, 337, 353, 379, 397, 401, 421, 433, 449, 461, 463, 491, 541, 547, 577, 599, 631, 673, 811,
829, 883, 1951, 3697.

These are 64 distinct primes; re-derive N by multiplying the list rather than trusting the printed decimal. Each factor is at most 3697, so its primality is certified by exact trial division (a deterministic check at this size). One verifies directly that $(p-1) \mid (N-1)$ for every p ; equivalently, writing $\lambda(N) = \text{lcm}_{p \mid N}(p-1) = 589416059232000$, one has $N \equiv 1 \pmod{\lambda(N)}$. Hence N is a squarefree Carmichael number with $\omega(N) = 64$, and $S_{64} \leq N$. Since $10^{147} \leq N < 10^{148}$, the number N has 148 digits. The certificate is verified by `ref/ref_carmichael.py::verify_certificate`, an independent, dependency-free routine (squarefreeness, distinctness, per-factor primality, and Korselt) sharing no code with the search that produced N .

Remark 1. N was found by an exchange meet-in-the-middle search at modulus $M = 3\lambda(N) = 1768248177696000$ (so $N \equiv 1 \pmod{M}$; M is the search modulus, *not* $\lambda(N)$ itself). It is an incumbent, not claimed minimal, and improves a previously reported 149-digit candidate. Its 148 digits are consistent with the known neighbours only in the following precise sense: Webster's N_{63} has 145 digits, hence $N_{63} < 10^{145} \leq S_{64}$, and the N_{65} candidate has 151 digits, hence exceeds N . This uses the proved interval and the stated digit lengths; it is not an appeal to any general monotonicity of S_k .

3 Lower bound

Fix $Y = 10^{145}$. We show no 64-factor Carmichael number lies below Y .

3.1 A certified prime universe

Lemma 2 (Cofactor lower bound). *There is an explicit constant*

$C = 1812140595092971804464959059612576188399713525008997620073689836850955613086867158799858934921$

(a 142-digit integer) such that for every Carmichael number n with $\omega(n) = 64$ and every prime factor $q \mid n$, the product of the remaining 63 factors satisfies $n/q \geq C$.

Proof (finite computation). Fix such an n and a prime factor q ; let $p_1 < \dots < p_{63}$ be the other factors, all odd (Lemma 1) and pairwise admissible. Let S be the set of these that are ≤ 97 ; there are finitely many internally admissible such S . For a fixed S , let $b_1(S) < b_2(S) < \dots$ be the primes > 97 that are *individually* admissible with every element of S (i.e. $p \nmid (b-1)$ for all $p \in S$). The

actual cofactor tail $r_1 < \dots < r_{63-|S|}$ of n consists of primes > 97 each admissible with S , so $r_i \geq b_i(S)$ for every i . Hence

$$n/q = \prod_{p \in S} p \cdot \prod_{i=1}^{63-|S|} r_i \geq F(S) := \left(\prod_{p \in S} p \right) \prod_{i=1}^{63-|S|} b_i(S) \geq \min_S F(S) = C.$$

Three points make this rigorous: (i) the $b_i(S)$ need not be pairwise admissible with one another, but ignoring those further constraints only decreases the product, so $F(S)$ is a genuine lower bound; (ii) the computation uses a finite prime table (odd primes $\leq 10^4$) and checks, for every enumerated S , that it contains enough eligible tail primes, so no smaller completion is truncated away; (iii) the cofactor n/q itself need not be Carmichael. The minimum is taken over all 407760 admissible S . Thus C is a certified lower bound on the cofactor product; it is not claimed to be the smallest possible admissible 63-factor product. $\square$

Corollary 1 (Universe). *Every 64-factor Carmichael number $n < Y$ has largest prime factor*

$$P^+(n) = q \leq \left\lfloor \frac{Y-1}{C} \right\rfloor = 5518.$$

In particular every prime factor of such an n lies among the 727 odd primes ≤ 5518 (the largest being 5507).

Proof. Take $q = P^+(n)$ the largest factor; then $n = q \cdot (n/q) \geq qC$ by Lemma 2, and $n < Y$ gives $q \leq (Y-1)/C$; evaluate the floor. Every other factor is smaller, hence also ≤ 5518 . $\square$

3.2 Exhaustion

Theorem 2 (Lower bound). *There is no Carmichael number $n < 10^{145}$ with $\omega(n) = 64$. Hence $S_{64} \geq 10^{145}$.*

Proof (finite exhaustion). By Corollary 1 every candidate is a product of 64 distinct primes drawn from the universe U of 727 odd primes ≤ 5518 , with product $< Y$. Enumerate all increasing 64-subsets of U with product $< Y$, pruning a partial choice only by two *necessary* conditions: (i) fewer than the required number of primes remain; and (ii) the current product times the 64 – h smallest still-available primes individually compatible with the h already chosen already reaches Y (an exact-integer $R_{\min}$ test). Pairwise admissibility (Lemma 1) is enforced during descent—once p is chosen, every later prime q with $p \mid (q-1)$ is discarded—which removes no genuine Carmichael candidate, since any real completion survives pairwise filtering and its remaining factors lie in the current available pool. At each complete 64-product, test Korselt.

The enumeration terminates with exactly 127092 increasing pairwise-admissible 64-subsets of U whose product is below Y (equivalently 127092 distinct products, by unique factorization), of which *none* is Carmichael. Therefore no 64-factor Carmichael number is smaller than 10^{145} . $\square$

3.3 Reproducibility

The exhaustion was carried out by *three separate enumeration implementations*, and the universe bound C was *independently recomputed*; all agree. Everything below is at repository revision f872e32 of [AlexanderEastwood/carmichael-frontier](#).

1. `s64_lower_bound_certificate.py` runs two enumerators: a binary include/exclude walk with an incrementally maintained $\text{lcm}(p_i - 1)$ Korselt test (327049 reference nodes), and an ordered-choice walk using per-prime divisibility, a separate prime generator, and no bitmasks (163525 nodes). Both report 127092 admissible products and 0 Carmichael. They use different prime generators, traversal logic, and Korselt predicates, while *sharing* the cofactor-bound computation and the results/hash reporting code; the identical order-sensitive leaf digest `sha256 = e05f62598eea784b3f60f025c6ccbf30700115ecbef42702e24154dcece32321` is therefore a reproducibility aid, not an additional independent correctness check.
2. `carmichael_prime_bound.py` independently recomputes C and the cap $P^+ \leq 5518$ from first principles. It does *not* enumerate products or establish the absence of Carmichael numbers; it certifies only the universe.
3. `s64_lower_bound_oracle_check.py` is a separate clean-room enumerator (its own prime generation and ordered traversal) that takes the Korselt verdict at every one of the 127092 leaves from the frozen oracle `ref/ref_carmichael.py::verify_certificate` rather than from its own arithmetic, again finding 0 Carmichael. It was validated against the oracle's brute-force `least_with_k` on small k : $k = 3 \rightarrow 561$, $k = 4 \rightarrow 41041$, $k = 5 \rightarrow 825265$.

Exact commands: `python3 s64_lower_bound_certificate.py`; `python3 carmichael_prime_bound.py --X 1e145 --pi`; `python3 s64_lower_bound_oracle_check.py`.

4 The interval

Combining Theorem 2 (lower) with §2 (upper) proves Theorem 1: $10^{145} \leq S_{64} \leq N < 10^{148}$, so S_{64} has 146, 147, or 148 digits, with remaining exclusion interval $[10^{145}, N)$. This is a global statement about S_{64} ; it is *not* a determination of S_{64} , which would require excluding every 64-factor Carmichael number in $[10^{145}, N)$.

Remark 2 (Strengthening to 147–148 digits). A complete exact- k preproduct search over all 64-factor products $\leq 10^{146}$ returns no Carmichael number, giving $10^{146} < S_{64} \leq N$, i.e. S_{64} has 147 or 148 digits. This is a different tool from the enumerators of §3.3 (a depth-first preproduct search with an exact $R_{\min}$ bound and the $P\lambda(P) > B$ arithmetic-progression switch), and it is far cheaper than the $\geq 3.89 \times 10^{10}$ admissible-product count below 10^{146} because it rejects prefixes without materialising each product (≈ 71 s, single core). *Verification tier*: the $S_{64} \geq 10^{145}$ bound of Theorem 2 is the triple-verified core; this strengthening currently rests on the search engine, corroborated by two independent switch strategies both returning “none” and by the same engine reproducing “none” at 10^{145} (a fourth independent confirmation of that bound) and reproducing $S_{10} \dots S_{29}$ against OEIS A006931. An independent admissibility certificate at 10^{146} is the outstanding follow-up; the full exclusion of $[10^{145}, N)$ that would prove $S_{64} = N$ remains a machine-scale endgame.

5 Transfer to S_{65}

The cofactor machinery transfers to $k = 65$ with no new exhaustion. Let

$$A_{64} = \min\{\prod_{i=1}^{64} p_i : p_1 < \dots < p_{64} \text{ odd primes, pairwise admissible}\}$$

be the least product of 64 pairwise-admissible odd primes. The enumeration of §3 records it as a by-product: it is the minimum over the 127092 admissible 64-products below 10^{145} (that this global minimum lies below 10^{145} is witnessed by those products), namely the explicit 145-digit integer

$$A_{64} = 441172365445881169056428109132100061485293342780359484110673067879903768586477795940804550206$$

Lemma 3 (Cofactor transfer). *For every Carmichael number n with $\omega(n) = 65$ and every prime $q \mid n$, the remaining 64 factors form a pairwise-admissible 64-product (Lemma 1), so $n/q \geq A_{64}$. Hence any such $n < U$ has $P^+(n) \leq \lfloor (U - 1)/A_{64} \rfloor$.*

Taking $U = 10^{148}$ gives $P^+ \leq 2266$: a universe of 334 odd primes.

Theorem 3 (Lower bound for S_{65}). *There is no Carmichael number $n < 10^{148}$ with $\omega(n) = 65$. Hence $S_{65} \geq 10^{148}$, and the smallest 65-factor Carmichael number has at least 149 digits.*

Proof (finite exhaustion). By Lemma 3 every candidate is a product of 65 distinct primes from the 334-prime universe with product $< 10^{148}$. An ordered-prefix search with exact product pruning descends until $P\lambda(P) > 10^{148}$ and then tests the at most one cofactor in the forced residue class $R \equiv P^{-1} \pmod{\lambda(P)}$, enforcing 65 distinct factors and full Korselt with primality taken from the sieve. It visits 64355 nodes, forms 66 progression candidates, and finds no Carmichael number. $\square$

Remark 3 (Reproduction and verification tier). Theorem 3 was reproduced four ways, all agreeing on 0 Carmichael numbers: (i) the certificate `s65/s65_exclusion.py` above; (ii) a separate native binary enumeration `s65/verify65.cpp` that checks all 22,287,138 admissible 65-products below 10^{148} without inverses or the progression shortcut; and (iii)–(iv) our preproduct search engine in both switch modes (§3.3), which independently reports $n = \text{none}$ with matching counts (5012 inversions, 66 progression candidates). A companion certificate `s65/s65_transfer_certificate.py` establishes the weaker $S_{65} \geq 3.17 \times 10^{147}$ by extending the recorded 64-products two independent ways (285 admissible 65-products, 0 Carmichael). All rest on the same A_{64} , whose defining 127092-product set and digest are the triple-verified core of §3.

Theorem 4 (Interval for S_{65}). $10^{148} \leq S_{65} \leq U_{65}$, where U_{65} is the 151-digit Carmichael number with exactly 65 prime factors 19, 31, 37, ..., 1021, 1993 (Webster’s candidate). Consequently S_{65} has 149, 150, or 151 decimal digits.

Proof. The lower bound is Theorem 3. For the upper bound: by Lemma 3 every 65-factor Carmichael below 10^{151} has all prime factors $\leq \lfloor (10^{151} - 1)/A_{64} \rfloor = 2266687$, so U_{65} (being $< 10^{151}$) factors completely over that range. Independent trial division confirms U_{65} is a product of exactly 65 distinct primes ≤ 1993 ; the frozen oracle `ref/ref_carmichael.py::verify_certificate` confirms Korselt, so U_{65} is Carmichael with $\omega = 65$ and $S_{65} \leq U_{65}$. As U_{65} has 151 digits and $S_{65} \geq 10^{148}$ has at least 149, the interval follows. (Factor list in `results_k65_webster_verified.json`.) $\square$

Acknowledgements

The known values for $3 \leq k \leq 63$ are due to R. G. E. Pinch (earlier k) and, for $k = 36$ –63, the Butler University computation; both are collected in J. Webster’s `small-carmichael-numbers`, the repository cited here for the known values and the N_{63} , N_{65} neighbours. Two existence results, of different scope, are relevant and should not be conflated: Alford, Grantham, Hayman, and Shallue constructed Carmichael numbers with exactly k prime factors for every $3 \leq k \leq 19,565,220$; Larsen and Wright proved that Carmichael numbers exist for every sufficiently large factor count. Neither determines a minimum; the contribution reported here is an explicit interval for the minimum S_{64} .

AI assistance (OpenAI GPT-6, the model designation recorded for the working session) contributed the lower-bound argument, generated the initial certificate code (`s64_lower_bound_certificate.py`), and executed the first computation. Independently of that script, the author recomputed the co-factor bound C (`carmichael_prime_bound.py`) and wrote a separate oracle-gated enumerator (`s64_lower_bound_oracle_check.py`); rerunning the supplied certificate is not counted as a separately implemented proof. The author retains responsibility for the mathematics and the code, and treats the model as an assistant rather than a mathematical authority.